

In economic and financial terms, **slippage** is a highly specific, practical example of a **transaction cost**. While transaction cost economics focuses on things like contracts and research, financial markets look at the exact mechanics of execution.

Here are the human-written study notes linking slippage to transaction costs.

## 1. What is Slippage?

**Slippage** is the difference between the *expected* price of a trade (the price you see on your screen when you click "buy" or "sell") and the *actual* executed price at which the trade goes through.

$$\text{Slippage} = \text{Actual Executed Price} - \text{Expected Price}$$

Slippage is considered an **implicit transaction cost**. Unlike explicit costs (such as a broker's flat \$5 fee or a tax), slippage is hidden inside the price movement itself, quietly eating into your returns. It can happen in any market—stocks, forex, commodities, and especially crypto.

## 2. Why Does Slippage Happen?

Slippage is caused by market friction during the short window of time between a trader submitting an order and that order being filled. It primarily happens due to two market conditions:

### A. Market Volatility

High-velocity price action moves faster than network latency. If you submit a market order to buy a stock at \$100.00, but the price jumps to \$100.25 in the milliseconds it takes your order to reach the exchange, your order fills at \$100.25. You just paid a 25-cent slippage cost per share.

### B. Low Liquidity (Market Depth)

Liquidity represents how many buyers and sellers are waiting at various price points (the order book). If you want to buy 10,000 shares of a thinly traded stock, there might only be 2,000 shares available at the current market price.

To fill your entire order, the broker has to walk "up the book," buying the remaining 8,000 shares at progressively higher prices.

## 3. Positive vs. Negative Slippage

Slippage isn't always bad, though it usually is for the party initiating a hasty market order.

- **Negative Slippage:** The price moves against you.
  - *Buying:* The execution price is **higher** than expected.

- *Selling*: The execution price is **lower** than expected.
- **Positive Slippage**: The price moves in your favor during the execution delay.
  - *Buying*: The execution price is **lower** than expected.
  - *Selling*: The execution price is **higher** than expected.

#### 4. Slippage in Decentralized Finance (DeFi)

In modern automated market makers (AMMs) like Uniswap, slippage takes on a unique mathematical form.

Because crypto AMMs rely on liquidity pools governed by formulas (like  $x \text{ times } y = k$ ), **every single trade shifts the price balance of the pool.**

- Small trades cause negligible slippage.
- Large trades relative to the total size of the pool cause massive, guaranteed negative slippage (often referred to as **Price Impact**).

#### 5. How Traders Manage Slippage Costs

Because slippage acts as a drag on portfolio performance, institutional and retail traders use specific tactics to mitigate it:

1. **Limit Orders**: Instead of using a market order (which says *"buy this right now at any price"*), traders use limit orders (*"only buy this if you can get it at \$100 or less"*). This completely eliminates negative slippage, though it introduces the risk that the order might never be filled.
2. **Slippage Tolerance Settings**: In crypto and algorithmic trading, users can set a "max slippage" percentage (e.g., 0.5%). If the price drifts further than 0.5% before the transaction settles, the smart contract or broker automatically cancels (reverts) the trade.
3. **Order Splitting (TWAP/VWAP)**: Institutional investors looking to buy millions of dollars of an asset do not execute it all at once. They use algorithms to chop the massive block into tiny orders spread across hours or days to avoid driving the market price up against themselves.